Unit-1

Mobile Communications:

- Introduction
- Guided transmission, unguided transmission- signal propagation frequencies, antennae, modulation, modulation methods and standards for voice- oriented data communication standards.
- Mobile computing- novel applications and limitations,
- Mobile computing architecture,
- Mobile system networks
- A simplified reference model.
- Mobile devices and systems- Cellular networks and frequency reuse
- Mobile smart phones
- Handheld pocket computers.

Introduction to Mobile Communication:

The rapidly expanding technology of cellular communication, wireless LANs, and satellite services will make information accessible anywhere and at any time. Regardless of size, most mobile computers will be equipped with a wireless connection to the fixed part of the network, and, perhaps, to other mobile computers.

Mobility and portability will create an entire new class of applications and, possibly, new massive markets combining personal computing and consumer electronics.

Mobile communication entails transmission of data to and from handheld devices. The location of the device can vary either locally or Globally.

Mobile Communication takes place through a wireless, distributed or diversified network and it is a two-way transmission or reception of data streams. Signals from a system can be transmitted through a fiber, wire, or wireless medium

Guided Transmission:

- Metal wires and optical fibres guided or wired transmission of data.
- Guided transmission of electrical signals takes place using four types of cables
  Optical fiber, Coaxial cable, Twisted-pair cable, Power line
- Fibre- and wire- based transmission and their ranges

Advantages:

- Transmission along a directed path from one point to another
- Practically no interference in transmission from any external source or path
- Using multiplexing and coding, a large number of signal-sources simultaneously transmitted along an optical fibre, a coaxial cable, or a twisted-pair cable

Disadvantages:

- Signal transmitter and receiver fixed.
- Number of transmitter and receiver systems limits the total number of interconnections possible

Unguided (Wireless) Transmission:

- Wireless or unguided transmission is carried out through radiated electromagnetic energy.
- Electromagnetic energy flows in free space (air or vacuum).
- The radiated energy is of frequency in MHz or GHz spectrum range.
- Spectrum means a set of frequencies in a range.

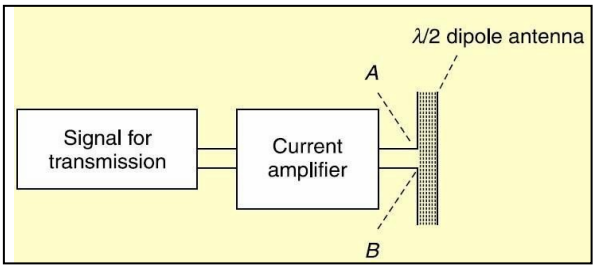
a) Signal Propagation Frequencies:

- Electrical signals transmitted by converting them into electromagnetic radiation.
- These radiations are transmitted via antennae that radiate electromagnetic signals.
- There are various frequency bands within the electromagnetic spectrum.
- The various types of frequencies are,
    - Long Wavelength (LW) radio, very low frequency.
    - Medium Wavelength (MW) radio, medium frequency.
    - Short Wavelength (SW) radio, high frequency.
    - FM radio band frequency.
    - Very High Frequency (VHF).

- Ultra High Frequency (UHF).
- Super High microwave Frequency (SHF).
- Extreme High Frequency (EHF).
- Far infrared
- Infrared
- Visible Light and Ultra-violet

b) Antennae:

- Devices that transmit and receive electromagnetic radiations.
- Most antennae function efficiently for relatively narrow frequency ranges.
- If an antenna not properly tuned to the frequency band in which the transmitting system connected to it operates, the transmitted or received signals may be impaired.



- The forms of antennae used are chiefly determined by the frequency ranges they operate in and can vary from a **single piece of wire** to a **parabolic dish.**

c) Modulation:

- Modulation means modification to original action so that the modification in the action is clearly presented.
- For example, a professor's voice is modulated and reflects his command over the subject.
- Similarly, electrical signals are modulated with information or electrical signals which is then communicated over long distances.
- Types of modulation are,
  - Analog signal modulation, Digital signal modulation, Amplitude modulation
  - Amplitude shift keying, Frequency modulation, Frequency shift keying
  - Phase modulation, Phase shift keying
  - Binary phase shift keying, Gaussian minimum shift keying,
  - Quadrature phase shift keying, Eight phase shift keying
  - Quadrature amplitude modulation (QAM).
  - 64-QAM

d) Modulation methods and standards for voice-oriented data communication standards:

1G: Devices have only voice-oriented communication.

2G: 2G devices communicate voice as well as data signals.

-Came onto the market in 1988.

-Support data rates up to 14.4 kbps.

2.5G and 2.5G+: -Support data rates up to 100 kbps.

3G:

-Higher data rates than 2G and 2.5G.

-Uses 2 Mbps or Higher for short distance transmission.

-Uses 384 Kbps for long distance transmission.

-Supports voice, data and multimedia streams.

-enable transfer of video clips and faster multimedia communication.

4G:

-Higher data rates than 3G.

-Support streaming data for video.

-Enables multimedia news paper, high resolution mobile TV.

-Support data rates up to 100 Mbps.

GSM and CDMA based standards and mobile communication network for long distance communication.

GSM: Global System for Mobile communication. It was developed by Groupe Speciale Mobile
(GSM) and Founded in Europe in 1982. It Support data rates up to 14.4 Kbps and Supports Cellular networks.

GSM900: It is using GMSK for transmitting 1's and 0's.

-Uses FDMA for channels and TDMA for user access in each deployed channel.

ii) EDGE and GPRS 2.5G and 3G:

-GSM has been enhanced to tri-band series and packet oriented data communication.
-GPRS is a packet-oriented service for data communication of mobile devices.
-Utilizes the unused channels in the TDMA mode in a GSM network.
-EDGE is an enhancement of the GSM phase 2.
-it has the data rates up to 48Kbps per 200KHz channel.

EGSM – Extended GSM.
GPRS – General Packet Radio Service.
EDGE – Enhanced Data rates for GSM Evolution.
EGPRS – Enhanced GPRS.

e) Modulation methods and standards for data and voice communication:

- CDMA – Code Division Multiple Access,
- FDMA – Frequency Division Multiple Access,
- TDMA – Time Division Multiple Access,
- WCDMA – Wireless CDMA
- UMTS – Upgraded WCDMA methods for downlink and uplink:
- High Speed Packet Data Access is provided by HSPDA and HSUPA.

Mobile Computing is a technology that allows transmission of data, voice and video via a computer or any other wireless enabled device without having to be connected to a fixed physical link.

Mobile Computing is also an umbrella term used to describe technologies that enable people to access network services anyplace, anytime, and anywhere.

A communication device can exhibit any one of the following characteristics:

- **Fixed and wired**: This configuration describes the typical desktop computer in an office. Neither weight nor power consumption of the devices allow for mobile usage. The devices use fixed networks for performance reasons.
- **Mobile and wired**: Many of today's laptops fall into this category; users carry the laptop from one hotel to the next, reconnecting to the company's network via the telephone network and a modem.
- **Fixed and wireless**: This mode is used for installing networks, e.g., in historical buildings to avoid damage by installing wires, or at trade shows to ensure fast network setup.
- **Mobile and wireless**: This is the most interesting case. No cable restricts the user, who can roam betweendifferent wireless networks. Most technologies discussed in this book dealwith this type of device and the networks supporting them. Today's most successful example for this category is GSM with more than 800 million users.

MOBILE COMPUTING ARCHITECTURE:

- It represents the architectural requirements for programming a mobile device.
- The requirements are Programming Languages, Functions of OS, and Functions of middleware for mobile systems.
- Mobile computing Architectural Layers, protocols and Layers

i) Programming Languages:

- A variety of programming languages are used in mobile computing architecture.
- Popular language used is Java
- J2ME and Javacard (Java for smartcard)
- J2EE is used for web and enterprise server-based applications of mobile services.
- DOTNET and Python 2.7 are also used.

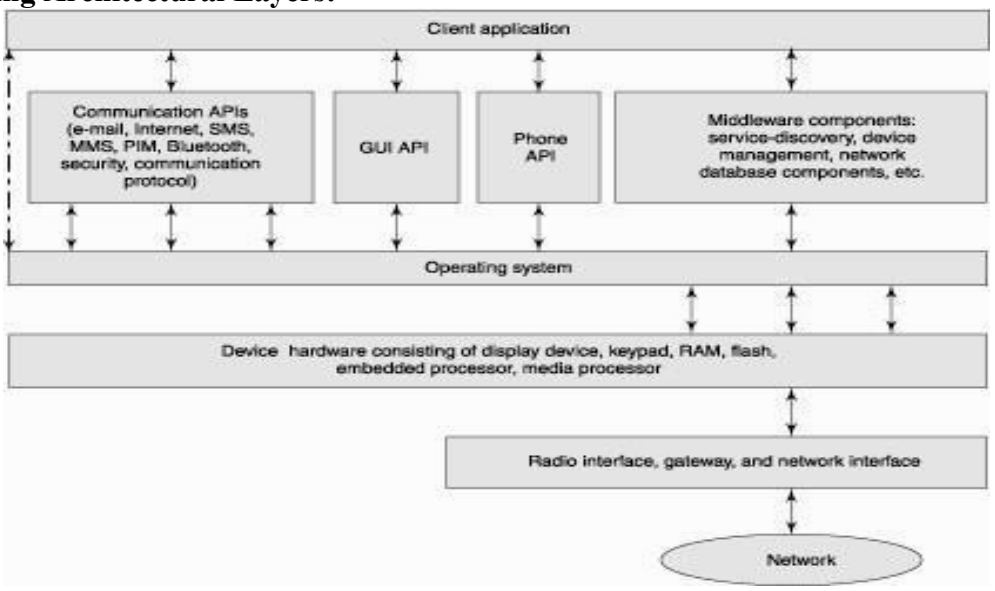
ii) Functions of OS:

- An OS enables the user to run an application without considering the hardware specifications and functionalities.
- Scheduling multiple tasks.
- Management functions for tasks and memory.
- Interfaces for communication.
- Configurable libraries for the GUI in the device.
- Middleware.

iii) Functions of middleware for mobile systems:

- Middleware are the software components that link the application components with the network-distributed components.
- Mobile OS also provides middleware components.
- Examples are
  - -to discover the nearby Bluetooth device, discover the nearby hotspot.
  - -to retrieve data from a network database.
  - -for service discovery

Mobile computing Architectural Layers:



- It refers to defining various layers between the user applications, interfaces, devices and network hardware.
- A well-defined architecture is necessary for systematic computations and access to data and software objects in the layers.

Protocols:

- GSM900, GSM900/1800/1900, CDMA, WCDMA, HSPA, UMTS, i-Mode, LTE, and WiMax.
- WPAN protocols such as Bluetooth, IrDA, and Zigbee.
- WLAN protocols such as 802.11a, and 802.11b and WAP

Layers: The OSI (open standard for interchange) seven-layer format is

-Physical for sending and receiving signals (TDMA or CDMA coding)
-Data link (Multiplexing), Networking (for linking to the destination)
-Wireless transport layer security (for establishing end-to-end connectivity)
-Wireless transaction protocol, Wireless session protocol, and Wireless application environment (for running a mob.appln., e.g., mobile e-business)

Novel Applications:

- A large number of applications are available.
- Recently made mobile TV realizable and developed ultra-mobile PC in march-2006
- Some applications include,

  - Smart-phones, Enterprise solutions, Music, Video and E-books, Mobile Cheque, Mobile Commerce, and Mobile based supply chain management.

SmartPhones:

- A Smartphone is a mobile phone with additional computing functions so as to enable multiple applications.
- For example, **Blackberry 8530 curve** has additional computational capabilities

-SMS, MMS, phone, email, address book, web browsing
-PIM software
-Integrated attachment viewing
-QWERTY style layout
-Send/End keys
-Bluetooth capability for hand-free talking via headset, ear buds, and car kits
-Speakerphone
-polyphonic ringtones for personalizing your device
-bright high resolution display, supporting over 65,000 colours
-WiFi 802.11b and WiFi 802.11g
-GPS Tracker
-Media voice or video or camera picture recording and communication
-Live TV
-MicroSD card 256MB

Enterprise Solutions:

- Enterprises or large business networks have huge database and documentation requirements.
- The term 'enterprise solutions' therefore refers to business solutions for corporations or enterprises.
- It includes specialized hardware or software programming for,

-Storage management
-Security
-Revision
-Distribution and so on.

Music, Video and E-books:

- The Apple iPods or iPhones or iPads have made it possible to listen one's favourite tunes anytime anywhere.
- iPads have made it possible to read one's favourite book anytime anywhere.
- Besides storing music these players can also be used to view photo albums, slide shows and video clips.

Mobile Cheque and Mobile Commerce:

- Mobile Cheque is a mobile based payment system employed during a purchase.
- The service is activated through text message exchanges between the customer, a designated retail outlet, and the mobile service provider.

- M-commerce is also a new trend, such as buying or selling of items through mobile internet between customers and organizations.
- Mobile devices also used for e-ticketing, i.e, for booking cinema, train, flight and bus tickets.

Mobile-based Supply Chain Management:

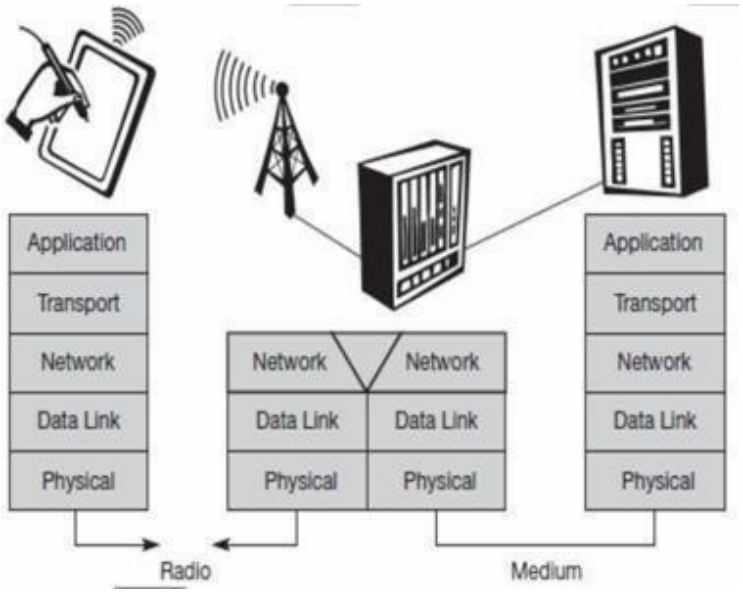
- The producer-consumer problem is called as SCM problem.
- Leading IT companies have developed mobile device software for SCM systems.
- The sales force and the manufacturing units use such mobile devices to maintain SCM.

Limitations of Mobile Computing:

- Resource Constraints : Battery
- Interference: *Radio transmission cannot be protected against interference using shielding and result in higher loss ratesfortransmitted data or higher bit error rates respectively.*
- Bandwidth: *Although they are continuously increasing, transmission rates are still very lowfor wireless devices compared to desktop systems. Researchers look for more efficient communication protocols with lowoverhead.*
- Dynamic changes in communication environment: *variations in signal power within a region, thus link delays and connection losses*
- Network issues : *discoveryof theconnection-servicetodestinationandconnectionstability*
- Interoperability issues*:*
- Security constraints: *Not only can portable devices be stolen more easily, but the radio interface is also prone to the dangers of eavesdropping. Wireless access must always include encryption, authentication, and other security mechanisms that must be efficient and simple to use.*

A simplified reference model

The figure shows the **protocol stack** implemented in the system according to the reference model. **End- systems** (such as the PDA and computer) need a full protocol to handle the application layer, transport layer, network layer, data link layer, and physical layer? Applications on the end-systems communicate with each other using the lower layer services. **Intermediate systems**, such as the interworking unit, do not necessarily need all of the layers.



A Simplified Reference Model

- **Physicallayer**: This is the lowest layer in a communication system and is responsible to convert the stream of bits into signals that can be transmitted on the sender side. The physical layer of the receiver then transforms the signals back into a bit stream. For wireless communication, the physical layer is responsible for frequency selection, generation of the carrier frequency, signal detection, modulation of data onto a carrier frequency and encryption.

- **Data link layer**: The main task of this layer is to access the medium, multiplexing of different data streams, correction of transmission errors, and synchronization (i.e., detection of a data frame). Therefore, the data link layer is responsible for a reliable point-to-point connection between two devices or a point-to-multipoint connection between one sender and several receivers.

- **Network layer**: This third layer is responsible for routing packets through a network or establishing a connection between two entities over many other intermediate systems. Important functions are addressing, routing, device location, and handover between different networks.
- **Transport layer**: This layer is used in the reference model to establish an end-to-end connection
- **Application layer**: Finally, the applications are situated on top of all transmission oriented layers. Functions are service location, support for multimedia applications, adaptive applications that can handle the large variations in transmission characteristics, and wireless access to the world-wide web using a portable device.